MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 07** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Periodic Test 07

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. The mean of first five prime numbers is: [1]

(A) 5.0 (B) 5.6 (C) 5.8 (D) 6.2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 5.6

Justification: The first five prime numbers are $2, 3, 5, 7, 11$.

$$\text{Mean} = \frac{2+3+5+7+11}{5} = \frac{28}{5} = \mathbf{5.6}.$$

$$\boxed{\text{Mean} = \mathbf{5.6}}$$

Q2. If $\sum f_i x_i = 1860$ and the mean of the distribution is 31, then the total frequency $\sum f_i$ is: [1]

(A) 50 (B) 55 (C) 60 (D) 65

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 60

Justification:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \implies 31 = \frac{1860}{\sum f_i} \implies \sum f_i = \frac{1860}{31} = \mathbf{60}.$$

$$\boxed{\sum f_i = \mathbf{60}}$$

Q3. In a frequency distribution, the class intervals are $1-10, 11-20, 21-30, 31-40$. The class size is: [1]

(A) 9 (B) 10 (C) 9.5 (D) 10.5

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 10

Justification: The true continuous boundaries for $1-10$ are $0.5-10.5$.

$$\text{Class size } (h) = 10.5 - 0.5 = \mathbf{10} \quad \text{(or difference between successive lower limits: } 11-1=10\text{)}.$$

$$\boxed{\text{Class Size} = \mathbf{10}}$$

Q4. The median of first 8 composite natural numbers is: [1]

(A) 9 (B) 9.5 (C) 10 (D) 8.5

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 9.5

Justification: The first 8 composite numbers arranged in ascending order are: $4, 6, 8, 9, 10, 12, 14, 15$.
Since $n = 8$ is even:

$$\text{Median} = \frac{4\text{th term} + 5\text{th term}}{2} = \frac{9+10}{2} = \mathbf{9.5}.$$

$$\boxed{\text{Median} = \mathbf{9.5}}$$

Q5. If the mean of a data is 27 and mode is 18, then the value of (Mean − Median) is: [1]

(A) 3 (B) 6 (C) 9 (D) 4.5

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 3

Justification: Using the empirical relationship:

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$$

Given Mean = 27 and Mode = 18:

$$27 - 18 = 3(\text{Mean} - \text{Median}) \implies 9 = 3(\text{Mean} - \text{Median}) \implies \text{Mean} - \text{Median} = \mathbf{3}.$$

$$\boxed{\text{Mean} - \text{Median} = \mathbf{3}}$$

Q6. **Assertion (A):** For a given distribution, $\sum_{i=1}^{n} f_i(x_i - \bar{x}) = 0$.
Reason (R): The sum of the products of frequencies and deviations of midpoints from the arithmetic mean is always zero. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification:

$$\sum f_i(x_i - \bar{x}) = \sum f_i x_i - \bar{x}\sum f_i = \sum f_i x_i - \left(\frac{\sum f_i x_i}{N}\right) N = 0.$$

Thus, the mathematical formulation in Assertion (A) is true, and Reason (R) accurately verbalizes and explains this fundamental algebraic property.

SECTION B: Very Short Answer Type

MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. Convert the following cumulative distribution into standard class intervals: [2]

Marks	Less than 10	Less than 20	Less than 30	Less than 40
Students	5	13	25	32

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Setting Up Class Intervals [1 Mark]

The cumulative frequencies indicate upper limits with class width of 10: $0-10, 10-20, 20-30, 30-40$.

Step 2: Computing Individual Frequencies [1 Mark]

Class Interval	Cumulative Frequency (cf)	Frequency (f_i)
$0-10$	5	5
$10-20$	13	$13-5=\mathbf{8}$
$20-30$	25	$25-13=\mathbf{12}$
$30-40$	32	$32-25=\mathbf{7}$

Q8. Find the class mark and class size of the interval $45-55$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Class Mark Calculation [1 Mark]

$$\text{Class Mark } (x_i) = \frac{\text{Upper Limit} + \text{Lower Limit}}{2} = \frac{55+45}{2} = \frac{100}{2} = \mathbf{50}.$$

Step 2: Class Size Calculation [1 Mark]

$$\text{Class Size } (h) = \text{Upper Limit} - \text{Lower Limit} = 55-45 = \mathbf{10}.$$

$$\boxed{\text{Class Mark} = \mathbf{50}, \quad \text{Class Size} = \mathbf{10}}$$

SECTION C: Short Answer Type

MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Compute the mode of the following distribution: [3]

Class	$0-5$	$5-10$	$10-15$	$15-20$	$20-25$	$25-30$
Frequency	4	8	18	12	6	2

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Modal Class & Parameter Identification [1 Mark]

The maximum frequency is 18, which corresponds to the class **10 – 15**.
$\therefore$ Modal Class $= 10 - 15$.
Lower limit $(l) = 10$, class width $(h) = 5$.
$f_1 = 18$ (modal frequency), $f_0 = 8$ (preceding frequency), $f_2 = 12$ (succeeding frequency).

Step 2: Formula Declaration & Substitution [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 10 + \left(\frac{18 - 8}{2(18) - 8 - 12}\right) \times 5 = 10 + \left(\frac{10}{36 - 20}\right) \times 5 = 10 + \left(\frac{10}{16}\right) \times 5$$

Step 3: Calculation & Final Result [1 Mark]

$$\text{Mode} = 10 + \frac{50}{16} = 10 + 3.125 = \mathbf{13.125} \quad \left(\text{or } 13\frac{1}{8}\right)$$

$$\boxed{\text{Mode} = \mathbf{13.125} \quad \left(\text{or } \mathbf{13\frac{1}{8}}\right)}$$

Q10. If the median of the following data is 35, find the missing frequency k: [3]

Class	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$	$50-60$
Frequency	2	3	k	6	5	3

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Cumulative Frequency & Median Class [1 Mark]

Given Median = 35, which lies in the class interval **30 – 40**.
$\therefore$ Median Class $= 30 - 40$.
Cumulative frequencies: $0-10 : 2$; $10-20 : 5$; $20-30 : 5+k$ (cf of preceding class); $30-40 : 11+k$; $40-50 : 16+k$; $50-60 : 19+k$.
Total frequency $N = 19 + k \implies \dfrac{N}{2} = \dfrac{19+k}{2}$.

Step 2: Median Formula & Equation Setup [1 Mark]

Here, $l = 30$, $f = 6$, $h = 10$, $cf = 5 + k$.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h \implies 35 = 30 + \left(\frac{\frac{19+k}{2} - (5+k)}{6}\right) \times 10$$

$$5 = \left(\frac{19 + k - 10 - 2k}{12}\right) \times 10 \implies 5 = \frac{5(9-k)}{6}$$

Step 3: Solving for k & Final Boxed Answer [1 Mark]

$$1 = \frac{9-k}{6} \implies 6 = 9 - k \implies k = 9 - 6 = 3.$$

$$\boxed{\text{Missing Frequency } k = \mathbf{3}}$$

SECTION D: Long Answer Type

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. The median of the following frequency distribution is 525. Find the values of x and y, if the total frequency is 100: **[5]**

Class Interval	Frequency	Class Interval	Frequency
$0-100$	2	$500-600$	20
$100-200$	5	$600-700$	y
$200-300$	x	$700-800$	9
$300-400$	12	$800-900$	7
$400-500$	17	$900-1000$	4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Cumulative Frequency Table & Equation (i) [1.5 Marks]

Class	f_i	cf	Class	f_i	cf
$0-100$	2	2	$500-600$	20	$56+x$
$100-200$	5	7	$600-700$	y	$56+x+y$
$200-300$	x	$7+x$	$700-800$	9	$65+x+y$
$300-400$	12	$19+x$	$800-900$	7	$72+x+y$
$400-500$	17	$36+x$	$900-1000$	4	$76+x+y$

Given total frequency $N = 100$:

$$76 + x + y = 100 \implies x + y = 100 - 76 \implies x + y = 24 \quad \text{— (i)}$$

Step 2: Median Class Identification & Parameter Values [1 Mark]

Given Median = 525, which lies in the class interval **500 – 600**.

$\therefore$ Median Class = $500 - 600$.

Lower limit $(l) = 500$, $f = 20$, $h = 100$, $\dfrac{N}{2} = 50$, $cf = 36 + x$ (preceding cf).

Step 3: Solving for x via Median Formula [1.5 Marks]

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$$

$$525 = 500 + \left(\frac{50 - (36 + x)}{20}\right) \times 100$$

$$25 = (14 - x) \times 5 \implies 5 = 14 - x \implies x = 14 - 5 = \mathbf{9}.$$

Step 4: Finding y & Final Result [1 Mark]

Substituting $x = 9$ into Equation (i):

$$9 + y = 24 \implies y = 24 - 9 = \mathbf{15}.$$

$$\boxed{x = \mathbf{9}, \quad y = \mathbf{15}}$$

Q12. The mean of the following distribution is 62.8 and the sum of all frequencies is 50. Find the missing frequencies f_1 and f_2: **[5]**

Class	$0-20$	$20-40$	$40-60$	$60-80$	$80-100$	$100-120$
Frequency	5	f_1	10	f_2	7	8

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Step 1: Total Frequency Equation [1 Mark]

$$\sum f_i = 5 + f_1 + 10 + f_2 + 7 + 8 = 30 + f_1 + f_2 = 50$$

$$f_1 + f_2 = 50 - 30 \implies f_1 + f_2 = 20 \quad \text{— (i)}$$

Step 2: Direct Method Calculation Table [1.5 Marks]

Class	Frequency (f_i)	Class Mark (x_i)	$f_i x_i$
$0 - 20$	5	10	50
$20 - 40$	f_1	30	$30f_1$
$40 - 60$	10	50	500
$60 - 80$	f_2	70	$70f_2$
$80 - 100$	7	90	630
$100 - 120$	8	110	880
Total	$\sum f_i = 50$		$\sum f_i x_i = 2060 + 30f_1 + 70f_2$

Step 3: Setting Up Mean Equation [1.5 Marks]

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \implies 62.8 = \frac{2060 + 30f_1 + 70f_2}{50}$$

$$62.8 \times 50 = 2060 + 30f_1 + 70f_2 \implies 3140 = 2060 + 30f_1 + 70f_2$$

$$30f_1 + 70f_2 = 3140 - 2060 \implies 30f_1 + 70f_2 = 1080$$

Dividing by 10:

$$3f_1 + 7f_2 = 108 \quad \text{— (ii)}$$

Step 4: Solving Simultaneous Equations [1 Mark]

Multiplying equation (i) by 3: $3f_1 + 3f_2 = 60$.
Subtracting from equation (ii):

$$(3f_1 + 7f_2) - (3f_1 + 3f_2) = 108 - 60 \implies 4f_2 = 48 \implies f_2 = \mathbf{12}.$$

From (i): $f_1 + 12 = 20 \implies f_1 = \mathbf{8}$.

$$\boxed{f_1 = \mathbf{8}, \quad f_2 = \mathbf{12}}$$

SECTION E: Case Study Based Question

MATHS MANTHAN | (1 × 4 = 4 Marks)

Q13. CASE STUDY: INDUSTRIAL MANUFACTURING PLANT DAILY WAGES AND WELFARE AUDIT [4 Marks]

A labor enforcement commissioner conducted a statutory wage survey of 50 skilled and semi-skilled factory workers in an industrial export zone.

Daily Wages (Rs)	$200 - 250$	$250 - 300$	$300 - 350$	$350 - 400$	$400 - 450$
Number of Workers	12	14	8	6	10

(i) Find the mean daily wage of the factory workers using the Assumed Mean Method. [2]

(ii) Identify the modal wage bracket of the workers. [1]

(iii) Find the number of workers earning less than Rs 350 per day. [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Mean Daily Wage by Assumed Mean Method [2 Marks]

Let Assumed Mean $a = 325$.

Wages (Rs)	Workers (f_i)	Class Mark (x_i)	$d_i = x_i - 325$	$f_i d_i$
$200 - 250$	12	225	-100	-1200
$250 - 300$	14	275	-50	-700
$300 - 350$	8	325	0	0
$350 - 400$	6	375	$+50$	$+300$
$400 - 450$	10	425	$+100$	$+1000$
Total	$\sum f_i = 50$			$\sum f_i d_i = -600$

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i} = 325 + \frac{-600}{50} = 325 - 12 = \mathbf{313} \text{ Rs.}$$

$$\boxed{\text{Mean Daily Wage} = \text{Rs } \mathbf{313}}$$

Subpart (ii): Modal Wage Bracket [1 Mark]

The highest frequency of workers is 14, which lies in the wage bracket $\mathbf{250 - 300}$.

$$\boxed{\text{Modal Wage Bracket} = \text{Rs } \mathbf{250 - 300}}$$

Subpart (iii): Workers Earning Less Than Rs 350 per Day [1 Mark]

Number of workers earning $<$ Rs 350 comprises classes $[200 - 250]$, $[250 - 300]$, and $[300 - 350]$:

$$\text{Total Workers} = 12 + 14 + 8 = \mathbf{34}.$$

$$\boxed{\text{Workers earning } < \text{Rs } 350 = \mathbf{34}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★